# Database Design Assignment 1

Due: Friday October 24th 2025
Worth: 20%

A Car hire firm, who are generally considered to be a shady company somehow earning large profits, have hired you at a great expense to set up a new database system for them.

They have requested that you show them how you would use your knowledge of Normalization to design their new database system. The table below represents a snapshot of their customer car hire data. Each customer may hire cars from various outlets throughout Cork. A car is registered at a particular outlet and can be hired by a customer on a given date.

You should answer all questions that follow, below the table. You should also state clearly any assumptions you are making with respect to the data and your interpretation of the attributes listed.

| **carReg** | **make** | **model** | **custNo** | **custName** | **hireDate** | **outletNo** | **outletLoc** |
|---|---|---|---|---|---|---|---|
| M565 0GD | Ford | Escort | C100 | Smith, J | 14/5/98 | 01 | Bandon |
| M565 0GD | Ford | Escort | C201 | Hen, P | 15/5/98 | 01 | Bandon |
| N743 TPR | Nissan | Sunny | C100 | Smith, J | 16/5/98 | 01 | Bandon |
| M134 BRP | Ford | Escort | C313 | Blatt, O | 14/5/98 | 02 | Mallow |
| M134 BRP | Ford | Escort | C100 | Smith, J | 20/5/98 | 02 | Mallow |
| M611 0PQ | Nissan | Sunny | C295 | Pen, T | 20/5/98 | 02 | Mallow |

**(a)** What stage of development is the above table currently in within the design process: UNF, 1NF, 2NF, 3NF or BCNF?
Explain your choice of answer.
*(5 Marks)*

**(b)**The data in the above table is susceptible to processing anomalies. Provide two examples for each of insertion, deletion, and update anomalies showing how they might occur in the above table.
*(15 Marks)*

**c)**Using the process of Normalization, carry out the transformation of the above table to BCNF. All the steps should be clearly shown and labelled; all keys and functional dependencies should be identified at each stage. Identify the primary and foreign keys in your final BCNF relations.
*(60 Marks)*

**(d)** Draw an Entity–Relationship model for the normalization process completed in step (c) above. Show all the entities, relationships, and attributes.
*(20 Marks)*